

Leave Approval System using ServiceNow - Capstone Project Documentation

1. Introduction

Project Title: Leave Approval System using ServiceNow

2. Project Overview

Purpose: To automate corporate leave requests, implement intelligent client-side date restrictions, and streamline managerial approval routing using the ServiceNow platform.

Features: Custom scoped application table, client-side input validation, automated manager approval routing via Workflow Studio (Flow Designer), multi-channel notifications (email and system UI badges), and operational dashboards.

The **Leave Approval System** is a ServiceNow-based automation solution designed to digitize and streamline the process of employee time-off requests, validation, and approval orchestration. The system eliminates manual tracking overhead, reduces HR processing delays, and provides transparency over organizational absence tracking.

The application is built within a secure scoped framework, utilizing a dedicated custom form interface where employees select their parameters, such as leave types (Annual Leave, Sick Leave, Casual Leave, or Unpaid Leave), start dates, end dates, and target manager. To maintain strict data integrity, the system utilizes front-end client script logic to enforce scheduling constraints, instantly preventing users from submitting an end date that is earlier than the start date.

Once a request is submitted, a record is written automatically to the system database. This record generation triggers background automation workflows via **Workflow Studio (Flow Designer)**. The workflow dynamically checks the assigned manager field and routes an electronic approval request to their workspace, pausing the request lifecycle until an action is executed.

Upon receiving the manager's decision, the platform triggers transactional status updates in real-time. Simultaneously, multi-channel notifications—including automated outbound emails and Next Experience notification bell alerts—are fired to the employee, ensuring clear and instantaneous feedback regarding their leave status.

Overall, the solution demonstrates how ServiceNow can scale beyond IT Service Management to construct tailored Line-of-Business workflows utilizing robust script validation, granular cloud orchestration, and unified event auditing.

3. Architecture

- **Presentation Layer (Frontend):** Custom Scoped App Form / Unified Navigation Banner
- **Application Logic Layer (Backend):** Client Scripts, Workflow Studio (Flow Designer) Processes
- **Data Layer (Database):** Custom Leave Request Table (x_snc_leave_app_leave_request), User Table (sys_user)

The **Leave Approval System** relies on a clear, multi-tiered ServiceNow architecture designed to separate end-user interactions, execution rules, database transactions, and data monitoring layers.

1. Presentation Layer (Frontend)

The user interaction boundary is formed by a dedicated scoped form interface. Employees access this form to provide key attributes such as their leave classification, duration windows, and required approvers. Client-side logic intercepts this layer using **onChange Client Scripts**, validating date configurations instantly before data hits the server to guarantee form accuracy and provide reactive UI feedback.

2. Application Logic Layer (Backend Processing)

The execution logic engine manages state changes and approvals through:

- **Client-Side Scripting:** Triggers immediately on input modifications to enforce relational business rules (e.g., clearing and alerting users about invalid end-date structures).
- **Workflow Studio (Flow Designer):** Houses the transactional lifecycle blocks. It initiates upon record insertion, maps the core approval routing gate, checks conditions, changes the record's underlying state field, and commands the distribution of communications.

3. Data Layer (Database)

Persistent information is maintained inside structured ServiceNow tables:

- **Leave Request Table (x_snc_leave_app_leave_request):** Tracks primary transaction schemas including Employee references, Leave Type choices, Start/End Dates, Status, and Manager references.
- **User Table (sys_user):** Out-of-the-box system directory containing employee metadata, organizational reporting paths, and target communication email endpoints.

4. Integration & Output Layer

When a state transition completes, the platform invokes its internal notification engines. It constructs outbound email elements via sys_email templates and pushes immediate UI updates directly to the employee's active navigation panel.

5. Reporting & Monitoring Layer

System monitoring dashboards utilize native analytics tracking tools to display critical workforce patterns:

- Total open vs. resolved leave requests.
- Distribution charts of absence categories.
- Departmental approval cycle velocities.

4. Setup Instructions

Prerequisites: ServiceNow Instance (Zurich / Xanadu Release)

Installation: Build a custom scoped application, construct tables, establish client scripts, and build approval workflows within Workflow Studio.

Step 1: Create Scoped Application & Data Table

- Navigate to **All > App Engine Studio** (or **System Applications > Studio**).
- Select **Create App** and name it **Leave Approval System**.

- Under the **Data** section, click **Add a table** and choose **Create from scratch**.
- Name the target table **Leave Request** (x_snc_leave_app_leave_request).
- Establish the baseline schema with the following custom columns:
 - Employee: Reference field pointing to the User (sys_user) table.
 - Leave Type: Choice list field containing: *Annual Leave, Sick Leave, Casual Leave, Unpaid Leave*.
 - Start Date: Date or Date/Time field.
 - End Date: Date or Date/Time field.
 - Status: Choice list field containing: *Requested, Approved, Rejected* (Set default value to *Requested*).
 - Manager: Reference field pointing to the User (sys_user) table.
 - Reason: Long String text area for business context.

Step 2: Configure Front-End Date Validation

- Navigate to **System Definition > Client Scripts** or add directly via Studio components.
- Configure a new **Client Script** with the following parameters:
 - *Name*: Validate Leave Dates
 - *Type*: onChange
 - *Field name*: end_date

Step 3: Build Automation Workflow in Workflow Studio

- Open **Workflow Studio** via **All > Process Automation > Workflow Studio** (or search for **Flow Designer**).
- Create a **New Flow** titled **Leave Approval Workflow**.
- Set the **Trigger** condition to run when a record is **Created** on the custom Leave Request table.
- Add **Action 1: Ask for Approval**:
 - *Target Record*: Select the triggering Leave Request Record.
 - *Approval Rule*: Choose **Approve when > Anyone approves**.
 - *Approver Mapping*: Use the data pill picker to select Trigger - Record Created > Leave Request Record > Manager.
- Add a **Flow Logic: If** block to check for approval completion:
 - *Condition*: Check Ask for Approval > Approval State is **Approved**.
 - *Sub-action A*: **Update Record** -> Set Status field to **Approved**.
 - *Sub-action B*: **Send Email** -> Send to Leave Request Record > Employee > Email with confirmation payload.

- *Sub-action C: **Send Notification*** -> Direct alert push to the employee's user account.
- Add a **Flow Logic: Else** block to catch rejections:
 - *Condition:* Check Ask for Approval > Approval State **is Rejected**.
 - *Sub-action A: **Update Record*** -> Set Status field to **Rejected**.
 - *Sub-action B: **Send Email*** -> Send to Leave Request Record > Employee > Email with rejection payload.

Step 4: System Verification and Testing

- Use the banner controls to **Impersonate User** and select a test employee profile.
- Open the custom form and attempt to specify an invalid entry where End Date occurs prior to Start Date to verify the operational UI alert.
- Submit a valid request form.
- End impersonation, return to the Admin dashboard, and navigate to **Workflow Studio > Executions** to check the status (*Waiting* status at the manager block).
- Impersonate the chosen manager user, navigate to their dashboard or approval task list, and select the authorization choice.
- Verify final execution state transitions, and check **System Logs > Emails** to validate output processing.

5. Folder / Component Structure

1. Form UI & Layout Layer

- Defines user entry boundaries for custom data fields (employee, leave_type, start_date, end_date, manager, reason).
- Manages end-user default parameter rendering and reference constraints.

2. Client-Side Scripting Components

- Enforces immediate frontend input parameters.
- Contains the onChange script architecture that blocks invalid date entries and sanitizes layout states.

3. Workflow & Logic Engine

- Consolidates runtime configurations inside ServiceNow's orchestration tools.
- Governs approval states, evaluation branches, field data manipulations, and system communication triggers.

4. Data Layer Architecture

- Maintains structural metadata definitions for the custom tables.
- Stores long-term historical request balances, process metrics, and transactional data references.

6. Running the Application

Access Form Interface → Fill Leave Request Parameters → Front-End Verification Enforced

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Notification Fired ←— Update Record State ←— Manager Approval ←— Flow Execution

- **Step 1:** The user loads the custom Leave Request input form layout.
- **Step 2:** The user enters the required dates and fields. The UI script runs instantly on changes to check date continuity rules.
- **Step 3:** The user submits the request record, writing it to the custom database and generating a *Requested* state stamp.
- **Step 4:** The automation engine detects record entry, evaluates user relationships, maps the workflow, and publishes a mandatory authorization event to the specified manager.
- **Step 5:** The system holds tracking states until the manager reviews and actions the task payload.
- **Step 6:** Upon action processing, the workflow updates the main status record and fires outbound notifications to the employee's devices and email queue.

7. API / Automation Logic

The application operates entirely within native ServiceNow engine rules, utilizing cloud-native platform hooks rather than complex external integration middleware.

Record Ingestion (Form)

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Trigger Flow Designer Logic

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Evaluate Approval Matrix Data Pills

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Halt & Await Manager Selection State

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|—► Approved —► Set Status: Approved —► Fire Outbound Events

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|—► Rejected —► Set Status: Rejected —► Fire Outbound Events

Data flows safely inside platform boundaries, resolving records via reference pills (Leave Request Record → Employee → Manager) to execute script instructions without exposed API endpoints.

8. Authentication & Access Security

Application integrity relies on ServiceNow’s native **Role-Based Access Control (RBAC)** architecture.

- **Authentication Hooks:** Managed entirely through core instance login gates. Sessions must be authorized before any workspace forms or flow elements can render.
- **Access Control Lists (ACLs):** Restrict system visibility at multiple layers:
 - *Read Access:* Configured so employees can exclusively track or review records where they are marked as the subject individual.
 - *Write Access:* Restricted to lock down request details once a workflow moves out of a *Requested* state.
 - *Manager Access:* Grants managers visibility to view and sign off on tasks assigned to them.
- **Server Protection Logic:** Ensures validation checks run securely in the background, preventing modifications to critical process statuses directly from the front-end layout.

9. User Interface

The interface focuses on functional simplicity to keep user data accurate and clear.

1. Employee Self-Service Layout

A clean form interface built with specific UI actions that automatically pre-fill user references upon load, providing clear, error-free date selection fields.

2. Request Status Portal

Provides a clear view for employees to check their submitted requests, displaying clear status indicators (*Requested*, *Approved*, or *Rejected*) along with tracking details.

3. Manager Approval Center

A dedicated dashboard workspace where supervisors can view outstanding leave requests, view reason text details, and quickly approve or reject requests.

10. Testing

1. Testcase Analysis Summary

Section	Total Cases	Not Tested	Fail	Pass
Leave Request Form UI	6	0	0	6
Date Validation Logic (Client Script)	8	0	0	8

Section	Total Cases	Not Tested	Fail	Pass
Workflow Studio Routing & Approval Gates	10	0	1	9
Status Field Automation States	5	0	0	5
Outbound Email Notifications Queue	6	0	1	5
In-App Platform Notifications (UI Bell Icon)	4	0	0	4

2. Evaluated Scenarios

- **Scenario 1 (Valid Submission):** Verifies that correct parameters transition cleanly into active workflow approvals. (*Pass*)
- **Scenario 2 (Date Enforcement):** Verifies that setting End Date < Start Date instantly blocks form submission and resets input variables. (*Pass*)
- **Scenario 3 (Dynamic Routing):** Verifies that requests route directly to the manager selected on the form. (*Pass*)

11. Known Issues & Limitations

1. **Static Balance Calculations:** The baseline app tracks approvals and states but does not maintain a live accrual balance counter (e.g., deducting accrued hours).
2. **Lack of Calendar Conflict Sweeper:** The system does not scan the team database to check if multiple team members are out on the same dates, leaving scheduling conflicts to manual manager review.
3. **Local Workspace Reliance:** Processing relies on users being connected to an active instance session; it does not support offline request queuing.

12. Future Enhancements

- **Dynamic Accrual Counter:** Building a background matrix system that deducts approved leave days directly from a user's total balance allocation.
- **Calendar Synchronization:** Integrating workflows with corporate calendars (like Outlook or Google Calendar) to automatically block out approved leave time.
- **Advanced Conflict Warning Engine:** Adding automated checks that flag a warning on the form if a significant percentage of the same department has already been granted leave during the requested dates.